

AlignAtt4LLM: Fast AlignAtt for Decoder-Only LLMs in an IWSLT 2026 Simultaneous Speech Translation System

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Abstract

We describe AlignAtt4LLM, an IWSLT 2026 simultaneous speech translation system for English-to-German, English-to-Italian, and English-to-Chinese. The system is a synchronous cascade: Qwen3-ASR with forced alignment produces an incrementally updated source transcript with word end times, and Gemma-4 E4B-it translates that prefix under an MT-side AlignAtt policy.

To our knowledge, this is the first application of AlignAtt to a decoder-only LLM, where the encoder-decoder cross-attention used by earlier AlignAtt systems is absent. We recover a usable policy through (1) an explicit source span in the causal prompt, (2) offline selection of translation-specific alignment heads, (3) selective *qk-fast* replay of the draft-to-source attention block, and (4) an engine-native query/key capture path that preserves model outputs bit-identically.

On the IWSLT 2026 development set, AlignAtt4LLM reaches a low-latency operating point near two seconds of compute-unaware LongYAAL and trades 1.32–1.53 additional seconds for clear gains in BLEU and XCOMET-XL in the high-latency regime. Head selection remains an end-to-end quality/latency trade-off, and the deployed *qk-fast* path keeps the MT observer cheap enough to run in the same serialized GPU loop as ASR.

1 Introduction

This paper describes the AlignAtt4LLM IWSLT 2026 submission system for English-to-German, English-to-Italian, and English-to-Chinese simultaneous speech translation. The system is a synchronous cascade: Qwen3-ASR with the Qwen3 forced aligner (Shi et al., 2026) produces an incrementally updated source transcript with word end times, and Gemma-4 E4B-it (Google DeepMind, 2026) translates the current source prefix under an MT-side AlignAtt policy. We keep two

latency-quality regimes, one tuned near the low-latency boundary and one tuned for higher translation quality.

The central contribution is to make an offline decoder-only LLM usable as an online MT module. Standard AlignAtt reads encoder-decoder cross-attention; Gemma-4 has no such cross-attention. We instead expose the source transcript as a known prompt span, select a small set of translation-specific self-attention heads offline, and accept only draft prefixes whose reconstructed attention signal remains within the currently available source frontier. This gives the offline Gemma translator an online policy without adding lexical repairs, phrase tables, or language-pair-specific streaming heuristics.

The second contribution is making that policy cheap enough for the submitted runtime. Both ASR and MT are served through vLLM (Kwon et al., 2023), which gives the cascade one high-throughput inference stack, CUDA-graph compilation, and hot model reuse across chunks. On the MT side, this also means the policy cannot inspect a Python-visible attention matrix: self-attention is hidden inside fused compiled kernels. We therefore capture the exact query/key tensors consumed by the deployed Gemma attention module and replay only the draft-to-source block needed by AlignAtt. The implementation will be released with the camera-ready version at <https://github.com/QuentinFuxa/Alignatt4LLM>.

Section 2 gives the task and prior-work context, Section 3 describes the submitted cascade, Section 4 details the MT-side AlignAtt realization, and Section 5 reports the evaluation.

2 Background

AlignAtt (Papi et al., 2023, 2024) is a simultaneous policy that lets an offline model operate in streaming mode. It has been used in

strong IWSLT simultaneous systems (Macháček and Polák, 2025). At each generation step, the policy reads an attention-derived source position for the next target token and stops generation when that position is not yet safely available. Previous AlignAtt systems obtained this signal from encoder-decoder cross-attention.

Decoder-only LLMs have recently become central in high-quality text-to-text MT systems (Kocmi et al., 2024, 2025). Prior simultaneous systems have already used instruction-tuned decoder-only LLMs by prompting them with the translation direction, current source prefix, and already emitted target prefix (Macháček and Polák, 2025). Long context, instruction following, and in-context examples make this family attractive for robust MT, but previous decoder-only simultaneous implementations did not use AlignAtt.

Translation-specific attention heads (Liu et al., 2026) show that multilingual decoder-only LLMs contain a sparse set of heads whose attention argmaxes track source-target word alignments. This suggests a decoder-only AlignAtt path, but only if two runtime conditions are met: the source span must be identifiable in the causal prompt, and the policy-visible attention rows must be reconstructed from the same tensors used by the deployed inference engine. The method below supplies both conditions.

2.1 IWSLT 2026 Simultaneous Task

Simulation environment. The IWSLT 2026 Simultaneous Task proposes the Simulstream toolkit (Gaido et al., 2025) for simulating simultaneous translation sessions. We follow the long-form, unsegmented task setup, which allows re-translation at the evaluation interface, but AlignAtt4LLM itself emits append-only incremental output.

Evaluation. We evaluate on the official IWSLT 2026 MCIF development set used in Section 5.1. Translation quality is reported with BLEU (Papineni et al., 2002), chrF (Popović, 2015), and XCOMET-XL (Guerreiro et al., 2024). Translation latency is reported with LongYAAL (Polák et al., 2025) in both compute-unaware and compute-aware variants.

3 AlignAtt4LLM System Overview

ASR. The speech side of the submitted cascade uses two upstream Qwen components: Qwen3-ASR-1.7B transcribes the live audio tail, and

Qwen3-ForcedAligner-0.6B assigns word-level start and end times to that transcript (Shi et al., 2026). The cascade invokes this combined Qwen path itself at every chunk boundary with timestamp output enabled; the word times are therefore produced online inside our simulation run, not added later as an offline post-processing step. Each request returns an updated transcript prefix together with per-word end times. We retain this dedicated aligner because it performed better than the Gemma AlignAtt ASR variants tested during development. Appendix A summarizes that comparison.

MT. The translation side uses Gemma-4 E4B-it (Google DeepMind, 2026) served through graph-compiled vLLM (Kwon et al., 2023). At step k , Gemma receives the current transcript prefix, the already accepted target prefix, and a fixed translation instruction, then greedily generates a draft continuation capped at 16 new tokens in the partial streaming mode used by the submitted bundles. AlignAtt accepts the longest draft prefix whose alignment signal stays on the currently accessible side of the source frontier.

Streaming loop. The cascade is synchronous and chunk-based. Each chunk first updates the ASR transcript prefix, then launches one MT request. Source words are considered accessible only once their end time is older than a hold-back margin; in the frozen submitted bundles this margin is 0 ms, so a word becomes accessible as soon as its aligned end time is observed. Both submitted regimes also delay the first MT emission until 2 seconds of source audio have accumulated. MT output is incremental-only: once a target token is accepted, it is reused as causal prefill for the next MT step and is never retracted.

Latency-quality parameters. We keep two official submission regimes. The low-latency submission uses $\Delta_{\text{chunk}} = 850$ ms and the high-latency submission uses $\Delta_{\text{chunk}} = 1500$ ms. Both use the same ASR backend, Gemma-4 MT backbone, and MT-side AlignAtt realization. The frozen submitted bundles use the same per-direction top-8 MT head sets, a width-7 source-axis median filter, and a border margin $b = 1$.

The submitted cascade itself is synchronous and single-GPU: each chunk triggers ASR first and MT second. This conservative regime cleanly separates policy behavior from scheduler overlap.

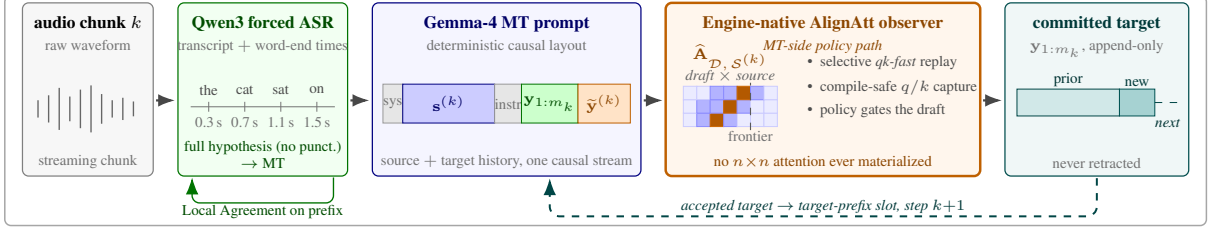


Figure 1: **Submitted cascade, one chunk-synchronous step.** Each chunk first updates the source prefix with Qwen3 forced ASR, then runs one Gemma-4 MT step. The MT prompt keeps the ASR transcript explicit inside the decoder-only causal layout, so the engine-native observer can capture the selected heads’ queries and keys on the deployed vLLM path and reconstruct only the draft-to-source block $\hat{\mathbf{A}}_{\mathcal{D},S^{(k)}}$ used by the acceptance policy. Accepted target tokens are appended to $\mathbf{y}_{1:m_k}$ and become the target-prefix slot of the next step’s prompt (dashed teal feedback).

Figure 2 summarizes the conceptual shift. The policy intuition remains the same as in encoder-decoder AlignAtt, but the alignment signal must now be recovered from a prompt-structured self-attention substrate rather than read directly from cross-attention.

4 Engine-Native AlignAtt for Decoder-Only LLMs

This section focuses on the method that enables AlignAtt for decoder-only LLMs.

4.1 Prompt Layout and Alignment Heads

Standard AlignAtt does not directly apply because a decoder-only LLM has no decoder cross-attention. We therefore make the source span explicit in the causal prompt. At chunk k , the serialized chat prompt is a concatenation of the system prompt, live transcript prefix, translation instruction, accepted translation prefix, and current draft:

$$\mathbf{p}^{(k)} = \left[\mathbf{p}^{\text{sys}} \parallel \mathbf{s}^{(k)} \parallel \mathbf{p}^{\text{instr}} \parallel \mathbf{y}_{1:m_k} \parallel \tilde{\mathbf{y}}^{(k)} \right], \quad (1)$$

where $\mathbf{s}^{(k)}$ is the live transcript prefix returned by ASR, $\mathbf{y}_{1:m_k}$ is the already accepted translation prefix, and $\tilde{\mathbf{y}}^{(k)}$ is the draft proposed by Gemma-4 at the current step. The transcript prefix is therefore a contiguous prompt span with a known map $\phi^{(k)}$ from source-word indices to prompt positions.

Not all self-attention heads carry a useful translation signal. Following Liu et al. (2026), we operationalize this with a two-stage offline calibration procedure under the exact prompt layout of Eq. (1): GPT-5-mini (OpenAI, 2025) first provides word-level source-target alignments on held-out parallel text, then every Gemma head is scored with Translation Score (TS) on those aligned examples. We retain the top $k = 8$ heads per language pair.

This head set is fixed at inference time and is the only part of the policy that depends on offline calibration. Appendix B reports the calibration details, corpus sizes, and per-direction score summaries. The retained MT heads are sparse and concentrated in a limited late-depth region of the backbone rather than being spread uniformly; Appendix B visualizes that pattern in Figure 8.

The policy only reads the source slice of each draft attention row, $\mathbf{A}_{t,\phi^{(k)}(s)}^{(\ell,h)}$, but unlike encoder-decoder cross-attention this slice is not source-normalized. Off-source mass on the accepted target prefix, prompt template, and speculative suffix is therefore structural rather than implementation noise. On the 78 text-only qualitative probes used for diagnostics, drafted units allocate on average 9% to accessible source tokens, 8% to inaccessible source tokens, 81% to non-source prompt positions, and 2% to the speculative suffix; among units that the policy actually accepts, the inaccessible-source share drops from 8% to 1%.

4.2 Selective Reconstruction for the Policy

The policy never needs the full $n \times n$ self-attention matrix. It only needs draft rows against source columns. For selected heads (ℓ, h) , draft positions t , and source positions s , we reconstruct

$$\hat{\mathbf{A}}_{t,s}^{(\ell,h)} = \frac{\exp\left(\gamma^{(\ell)} q_t^{(\ell,h)} \cdot k_{\phi^{(k)}(s)}^{(\ell,h)} + m_{t,\phi^{(k)}(s)}\right)}{\sum_j \exp\left(\gamma^{(\ell)} q_t^{(\ell,h)} \cdot k_j^{(\ell,h)} + m_{t,j}\right)}, \quad (2)$$

where q and k are the captured post-normalization, post-RoPE tensors actually consumed by the deployed attention module, $\gamma^{(\ell)}$ is that module’s runtime scaling factor, and $m_{t,j}$ replays the same causal and sliding-window masks as the fused forward. We call this replay path *qk-fast*: it recon-

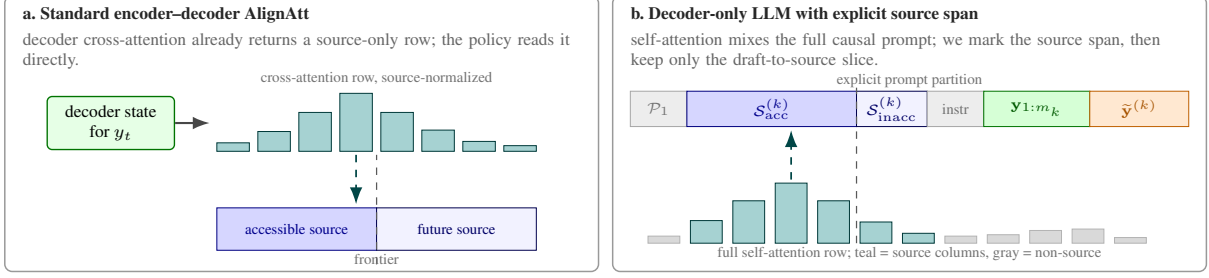


Figure 2: **How AlignAtt changes substrate between encoder-decoder and decoder-only models.** (a) In the original encoder-decoder setting, the decoder already exposes a source-only cross-attention row, so the policy can gate tokens directly against the accessible source frontier: if the peak of the row falls on the accessible side, the draft token is accepted. (b) In our decoder-only MT setting, source and target history share one causal prompt. We therefore mark the source span explicitly in prompt space, isolate translation-specific heads, and reconstruct only the draft-to-source slice needed by the policy; the decision rule is otherwise the same as in (a).

structs only the policy-visible block and matches the deployed attention algebra up to floating-point reassociation relative to the fused forward.

From these per-head rows we build two source-side provenance features for each drafted token: accessible-source mass π_t^{acc} and inaccessible-source mass π_t^{inacc} ,

$$\pi_t^{\text{acc}} = \sum_{s \in \mathcal{S}_{\text{acc}}^{(k)}} \bar{\mathbf{r}}_t(s), \quad (3)$$

$$\pi_t^{\text{inacc}} = \sum_{s \in \mathcal{S}_{\text{inacc}}^{(k)}} \bar{\mathbf{r}}_t(s), \quad (4)$$

where $\bar{\mathbf{r}}_t$ is the head-averaged reconstructed row. We additionally normalize per-head rows online with prefix Welford statistics and apply a short median filter along the source axis before taking the source-side argmax. The submitted gate only thresholds π_t^{acc} ; the complementary π_t^{inacc} is retained as a diagnostic split for Figure 9 and the appendix-level provenance analyses. This stabilizes the alignment peak without changing the underlying replayed rows.

4.3 AlignAtt Acceptance Policy

The submitted MT policy is a first-failure scan over drafted tokens. Let $\hat{s}_t$ be the source-side argmax of the filtered row for draft token t , and let $N_{\text{acc}}^{(k)}$ be the number of source words whose ASR end time is already on the accessible side of the frontier. The scan stops when one of three conditions first fires:

$$\text{SOURCE-FRONTIER : } \hat{s}_t \geq N_{\text{acc}}^{(k)} + b, \quad (5)$$

$$\text{ARGMAX-MASS-WEAK : } \bar{\mathbf{r}}_t(\hat{s}_t) < \tau_{\text{argmax}}, \quad (6)$$

$$\text{PROVENANCE-WEAK : } \pi_t^{\text{acc}} < \tau_{\text{src}}. \quad (7)$$

In the frozen 850/1500 submission bundles, $b = 1$, $\tau_{\text{argmax}} = 0$, and $\tau_{\text{src}} = 0$, so the optional argmax-mass and minimum-source-mass gates are present in the runtime but inactive in the submitted operating points.

Figure 4 makes the decision path explicit. One branch averages the retained heads and sums accessible-source mass to obtain π_t^{acc} . The other normalizes and smooths the same replayed rows before taking the source-side argmax $\hat{s}_t$. The gate accepts token t if the three tests in Eqs. (5)–(7) all pass; a left-to-right first-failure scan then emits the longest accepting draft prefix and rounds back to the last completed stability unit. Here a stability unit is the smallest target fragment that the tokenizer treats as safe to commit: a whitespace-delimited lexical word in spacing scripts such as EN→DE/IT, or a single CJK character in EN→ZH, so partial subword fragments are never emitted.

4.4 Engine-Native Capture under Compiled Inference

Equation (2) is only useful if the deployed runtime exposes the exact queries and keys consumed by attention. Under fused, graph-compiled vLLM, those tensors never appear as ordinary Python-visible objects. We therefore install a per-layer observer before compilation, copy prompt keys and draft-row queries into fixed-shape buffers, and route capture through a custom-op-style call that survives graph lowering. A zero-valued sentinel dependency keeps the observer in the transitive fan-in of the graph output:

$$\mathbf{o}'_\ell = \mathbf{o}_\ell + \Phi(\ell, \text{pos}, \mathbf{Q}^{(\ell)}, \mathbf{K}^{(\ell)}), \quad \Phi \equiv \mathbf{0}. \quad (8)$$

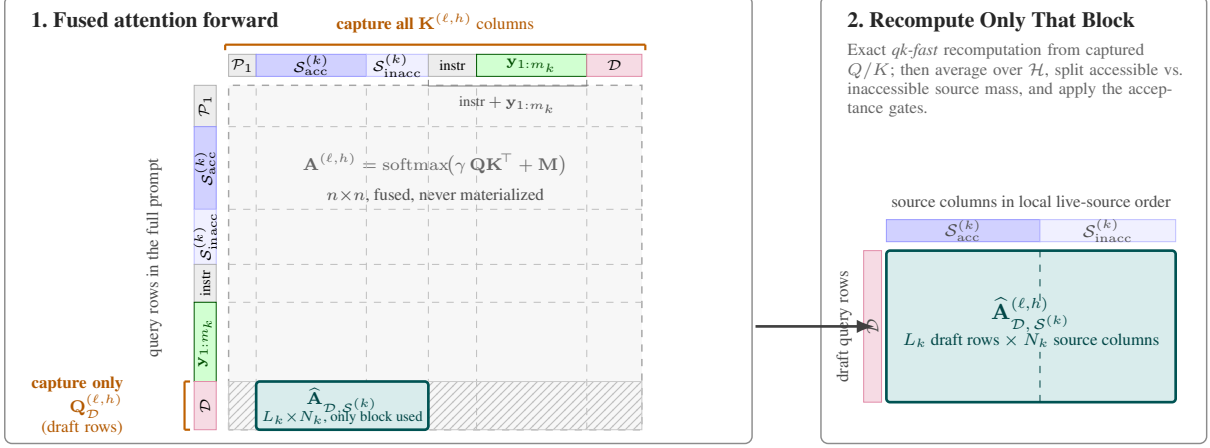


Figure 3: **Selective reconstruction with engine-native capture.** The $n \times n$ attention matrix $\mathbf{A}^{(\ell,h)}$ is executed entirely inside the fused attention kernel, so the full $n \times n$ matrix is never materialized. For each AlignAtt head $(\ell, h) \in \mathcal{H}$, the observer copies $\mathbf{K}^{(\ell,h)}$ for all positions and $\mathbf{Q}^{(\ell,h)}$ only for the draft rows $\mathcal{D}$ into fixed-shape buffers. The green band $y_{1:m_k}$ marks the committed target prefix that grows across chunks; hatched cells are never reconstructed. After the forward, Eq. (2) recomputes exactly only the draft-to-source block $\hat{\mathbf{A}}_{\mathcal{D},S^{(k)}}^{(\ell,h)}$, which is the sole object consumed by the source-side provenance split of Eqs. (3)–(4), where π_t^{acc} drives the gate and π_t^{inacc} is retained for diagnostics, and by the acceptance policy of this section.

Figure 5 separates the capture path into three stages. At compile time, we patch the attention forward and install fixed-shape observer slots before graph capture. During each compiled forward, the live observer records prompt keys and draft-row queries for the selected heads after the module’s q/k normalization and rotary embedding, then returns $\Phi \equiv \mathbf{0}$ back into the attention output. After the forward, a short Python-side replay reconstructs only the selective draft-to-source block consumed by the policy and by the parity suite. The sentinel add-back of Eq. (8) is therefore only a live-ness device: without the additive zero, graph lowering dead-code-eliminates the observer path and no usable Q/K buffers survive on the deployed runtime. Appendix C keeps the replay equations, parity measurements, and qualitative probe.

5 Results

5.1 Experimental Setup

All experiments run on a single NVIDIA A40. The MT backend uses Gemma-4 E4B-it in bfloat16 through graph-compiled vLLM; the speech side uses Qwen3-ASR with forced alignment. In evaluation, we use the official IWSLT 2026 MCIF dev set (~ 2.1 hours total, 21 long academic talks) resegmented by OmniSTEval v0.1.7. We report BLEU and chrF against the dev references, XCOMET-XL for translation adequacy (Guerreiro et al., 2024), and compute-unaware and aware

LongYAAL for latency (Polák et al., 2025). The streaming loop follows the Simulstream setting of unsegmented long-form audio with revision-capable evaluation (Gaido et al., 2025), but our submitted cascade itself is synchronous: ASR and MT are serialized on one GPU so that policy effects are not confounded with scheduler overlap.

For the auxiliary runtime study in Section 5.4, we compare three MT seams on a fixed suite of 16 text-only prompts: a minimal Transformers eager reference, a Transformers SDPA – *qk-fast* replay reference, and the deployed vLLM *qk-fast* path. Each seam uses greedy decoding with the same head set, one warmup pass, and repeated hot runs in isolated subprocesses.

5.2 Submitted System Results

Table 1 reports the submitted operating points. The pattern is consistent across EN→DE, EN→ZH, and EN→IT: the low-latency regime stays near a 2 s CU-LongYAAL point, while the high-latency regime moves to a clearly stronger quality point at the expected latency cost. CA-LongYAAL is consistently below CU-LongYAAL here because OmniSTEval’s computation-aware mode replaces each chunk-boundary audio increment by the actual wall-clock increment spent processing that chunk. Since the hot A40 system runs faster than real time, the computation-aware timestamps can be earlier than the compute-unaware chunk-boundary timestamps. The

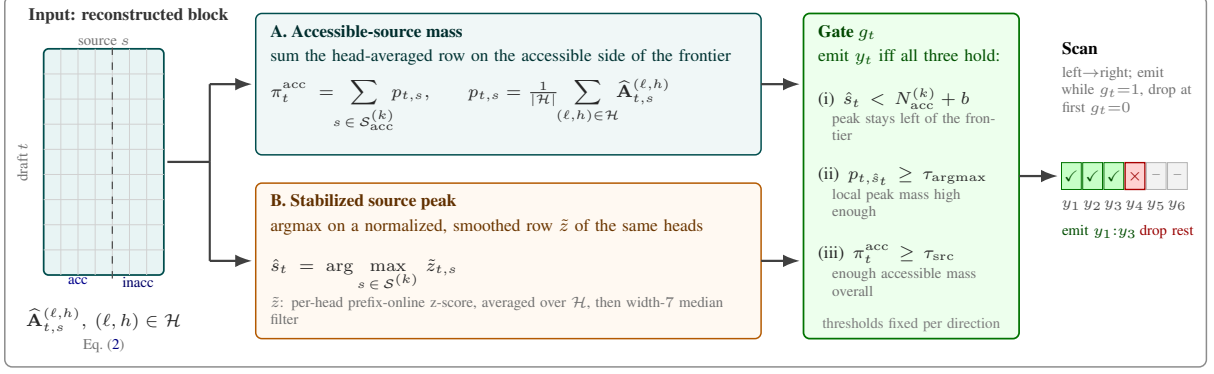


Figure 4: **From the reconstructed block to an acceptance decision.** The selective *qk-fast* reconstruction produces $\hat{A}_{t,s}^{(\ell,h)}$ for $(\ell, h) \in \mathcal{H}$, with draft positions t against source positions s (left). The policy reads it through two parallel aggregations. *Branch A* averages the selected heads into a row $p_{t,\cdot}$, and sums its mass on the accessible side of the frontier, giving the provenance score π_t^{acc} . *Branch B* returns the peak $\hat{s}_t$ of a stabilized row $\tilde{z}_{t,\cdot}$, where $\tilde{z}$ is obtained by z-scoring each head with prefix-online Welford moments, averaging over $\mathcal{H}$, and applying a width-7 median filter along the source axis. The gate g_t accepts draft token t iff the peak stays on the accessible side of the frontier, the peak mass exceeds τ_{argmax} , and the accessible-source mass exceeds τ_{src} . A left-to-right scan emits the longest run of accepting tokens and drops the remainder at the first failure.

Pair	Regime	BLEU	chrF	XCOMET-XL	LongYAAL (s)	
					CU	CA
en→de	low	28.76	62.1	0.875	2.00	1.63
	high	32.63	64.2	0.902	3.53	3.14
en→zh	low	36.01	35.0	0.743	1.95	1.77
	high	39.86	37.8	0.778	3.27	3.09
en→it	low	40.10	68.0	0.805	1.98	1.62
	high	44.46	70.1	0.841	3.48	3.10

Table 1: **Submitted cascade on the IWSLT 2026 dev set** (21 clips, OmniSTEval long-form resegmentation). Low rows use the submitted setting with $\Delta_{\text{chunk}} = 850$ ms. High rows use the submitted setting with $\Delta_{\text{chunk}} = 1500$ ms.

columns are therefore not inverted. We read Table 1 as evidence that the AlignAtt4LLM realization supports both a stable low-latency operating point and a higher-quality regime on the same decoder-only Gemma backend.

5.3 Head Filtering Matters End-to-End

Table 2 is an auxiliary EN→DE rerun included only to isolate the MT head-set effect under the maintained *qk-fast* runtime. End-to-end quality is nearly equivalent, but the latency split is informative. CU-LongYAAL increases by 100.3 ms, which likely means the all-head observer is slightly less precise and therefore a bit more generous on tokens whose support extends beyond the currently available source prefix. CA-LongYAAL rises more strongly, by +179.5 ms, because replaying all 336 heads forces the runtime to

Setting	XCOMET-XL ↑	CU (s) ↓	CA (s) ↓
Top-8 heads	0.879	1.96	1.65
All 336 heads	0.885	2.06	1.83

Table 2: **Auxiliary EN→DE MT head-set comparison at $\Delta_{\text{chunk}} = 1100$ ms.** Both rows use the same maintained runtime; only the MT head set changes. End-to-end quality is nearly equivalent, but all-head replay increases CU and especially CA because the observer becomes less selective while the runtime must reconstruct much more attention at each MT step.

reconstruct much more attention state at each MT step. We therefore read the auxiliary comparison as evidence that a small retained head set captures most of the useful MT alignment signal at much lower runtime cost. Appendix B gives the held-out calibration numbers behind the retained head sets.

5.4 Runtime Cost of the MT Observer

The MT observer must be cheap enough to justify deployment. Figure 6 shows that the engine-native vLLM replay path is much cheaper than the minimal eager seam we use as an inspection reference: median cost drops from 63.7 to 25.4 ms/token, while the Transformers SDPA – *qk-fast* replay remains close to the eager baseline at 59.0 ms/token. This confirms that the MT contribution is not merely inspectable in principle; it is practical inside the compiled runtime we actually deploy.

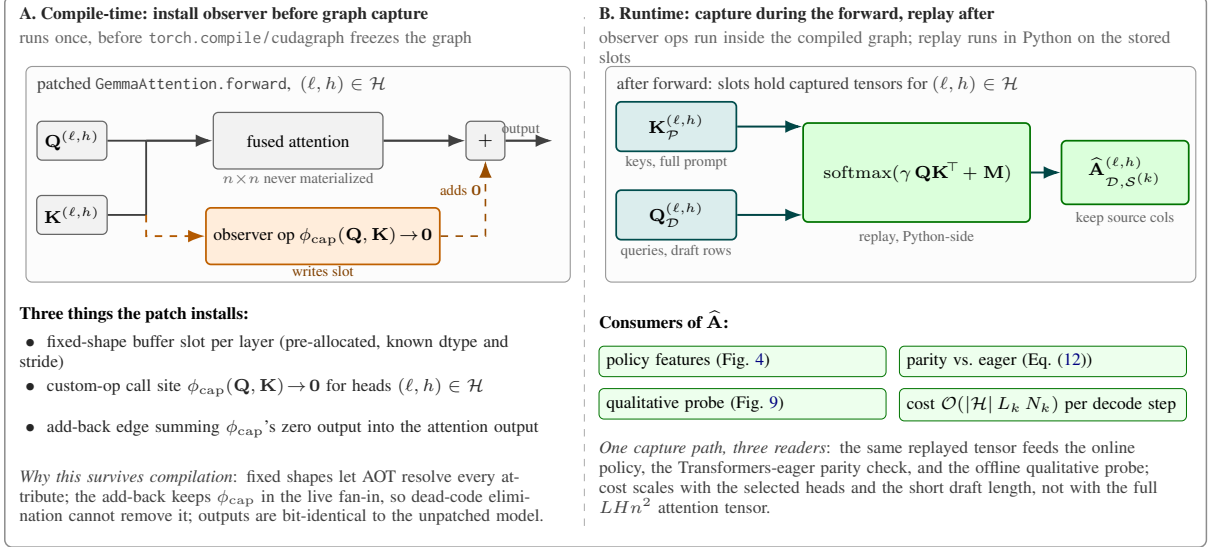


Figure 5: **Compile-safe observer lifecycle on the deployed vLLM path.** *Left (A), once, before graph capture:* the forward of `GemmaAttention` is patched so that for each selected head $(\ell, h) \in \mathcal{H}$ the query and key tensors also pass through an observer custom op ϕ_{cap} (dashed orange side path). The op writes the selected slices into fixed-shape, pre-allocated slots and returns a zero tensor that is added back into the attention output; this keeps the op in the live fan-in of the graph so the compiler cannot dead-code-eliminate it, while the numerical output of the layer is unchanged. *Right (B), every forward:* after the compiled forward finishes, the observer slots hold $K_P^{(\ell, h)}$ for the full prompt and $Q_D^{(\ell, h)}$ for the draft rows; a short Python-side replay reapplies the module scaling and the same prompt/suffix masks used by the deployed attention kernel, then keeps the source columns to yield the selective block $\hat{A}_{D, S^{(k)}}^{(\ell, h)}$ that feeds the policy, the parity check against an uncompiled Transformers reference, and the qualitative probe.

5.5 Coordination Regimes and Compute-Aware Latency

Our cascade loads the ASR and MT models as two separate blocking `vllm.LLM` instances on a single GPU, which serializes every audio chunk into an ASR decode followed by an MT decode (Fig. 7(c)). This conservative regime keeps policy effects separate from scheduler overlap. Alternative pipelined or fully asynchronous regimes would mainly reduce compute-aware LongYAAL at fixed compute-unaware latency, but would require a non-blocking scheduler and a more careful treatment of stale MT requests. We leave that scheduler axis to future work.

6 Conclusion

The retained IWSLT 2026 system is a Qwen3 forced ASR + Gemma-4 MT cascade, and its key technical ingredient is an engine-native implementation of AlignAtt for decoder-only MT under compiled inference. Deterministic prompt layout, offline head selection, selective *qk-fast* replay, and compile-safe query/key capture make the MT policy usable without leaving the deployed runtime.

On the dev set, this yields a low-latency operating point near 2 s CU-LongYAAL and a high-latency regime that is roughly 1.5 s slower but clearly stronger in BLEU and XCOMET-XL.

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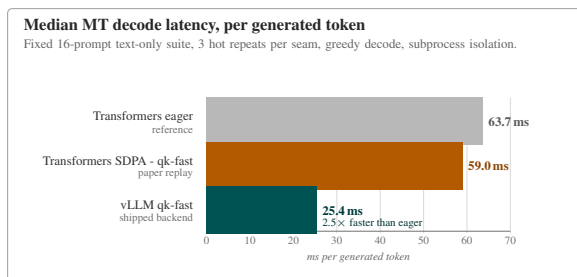


Figure 6: **Inference-time comparison of MT capture seams.** Median latency per generated token on a fixed 16-prompt text-only suite, for a minimal Transformers eager reference, a Transformers SDPA – *qk-fast* reference that reconstructs source rows from captured layer inputs, and the engine-native vLLM *qk-fast* path used by the shipped backend. Each seam uses one unmeasured warmup pass followed by three hot repeats, greedy decode, a fixed AlignAtt head set, and subprocess isolation; per-prompt results and divergences are stored in the generated JSON artifact.

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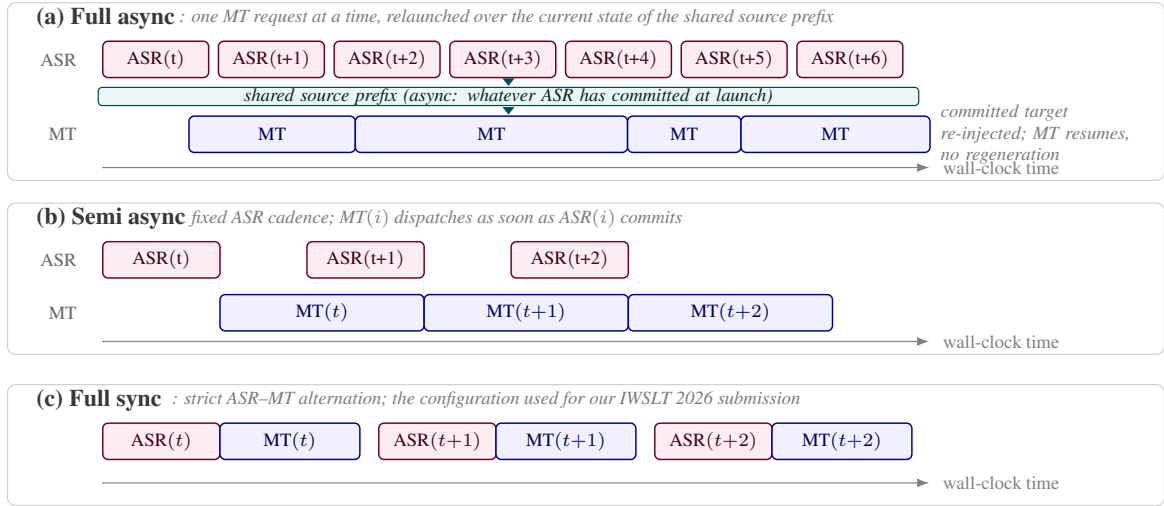
A Additional ASR Analyses

The analyses below summarize the all-Gemma ASR results that informed the final system choice but are no longer central to the retained submission narrative.

Gemma AlignAtt ASR is feasible but not competitive. We verified that the same AlignAtt toolkit can be made to run on the ASR side, but the resulting Gemma-ASR path is not competitive enough for submission. The limiting factor is poor Gemma-4 ASR quality itself, and this weakness appears to be shared by the current multimodal family rather than by one particular size variant: the same qualitative limitation is present for both E2B and E4B. On the diagnostic clip

ccpXHNfaoy (360 s), Qwen3 forced clearly dominates the exploratory Gemma AlignAtt ASR path: WER drops from 0.269 to 0.113, CER from 0.122 to 0.046, mean absolute boundary lag from 0.403 s to 0.145 s, load time from 123.0 s to 63.7 s, and wall-clock RTF from 0.628 to 0.306. We therefore keep Gemma AlignAtt ASR only as an exploratory result and retain Qwen3 forced ASR for the submitted system. If a future multimodal LLM closes that ASR gap, the same overall design could support a double-AlignAtt cascade in which AlignAtt governs both the ASR stage and the MT stage.

B Coordination Regimes and Additional MT Alignment-Head Diagnostics



Regimes (a) and (b) require a non-blocking scheduler (e.g. the vLLM async engine (Kwon et al., 2023)) with request abort on stale MT decodes; our current implementation is (c) with two blocking LLM instances sharing a single GPU. The gaps between ASR chunks in (a) and (b) reflect an ASR real-time factor below 1. This design axis mainly affects *compute-aware* latency; our central submitted comparisons remain on the serialized regime in (c).

Figure 7: Three coordination regimes for ASR and MT sharing one GPU. The submitted system uses the strict ASR-MT alternation in (c). In full async mode (a), successive MT launches need not correspond to comparable source-time increments: a long MT decode can absorb several incoming ASR updates, while the next relaunch may cover much less newly available audio. That makes history-dependent policies such as local agreement (Liu et al., 2020) less natural to calibrate there than an attention-only policy such as AlignAtt, which scores each draft directly against the currently accessible source prefix.

Figure 8 shows the architecture-aware score map behind the retained MT head set: useful heads are sparse, clustered late in the backbone, and overlap the late shared-KV region where observer cost is governed by KV ownership rather than raw query-head count. Table 3 then reports the held-out calibration gap between the retained top-8 heads and uniform all-head averaging.

C Observer Replay and Qualitative Diagnostics

Below we report the prompt/suffix replay equations, one qualitative probe, and the numerical parity measurements on the engine-native path.

Prompt-space replay details. After capture, we replay prompt-side and suffix-side logits separately from the stored buffers. Writing $\mathbf{B}_\star^{(\ell, h)} := \mathbf{B}_\star^{(\ell)}[h]$ for head-specific captured tensors, $\gamma^{(\ell)}$ for the attention module’s runtime scaling factor, and $\mathbf{M}_{\text{prompt}}^{(\ell, h)} / \mathbf{M}_{\text{suffix}}^{(\ell, h)}$ for the replayed prompt-side

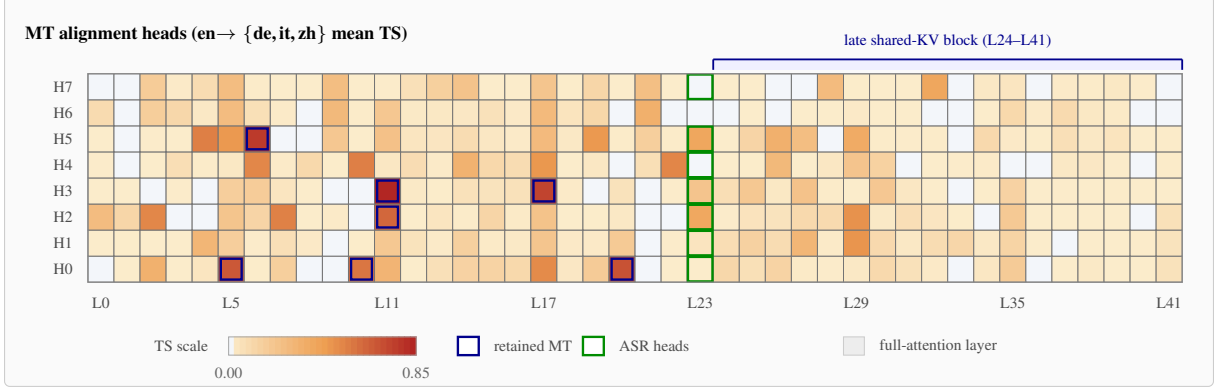


Figure 8: **Architecture-aware view of retained MT alignment heads.** The useful MT heads are sparse, late, and only partly overlap with the ASR set, which highlights role-specific specialization. Their clustering inside the late shared-KV block is favorable because observer cost follows KV-group ownership rather than raw query-head count.

Pair	Top-8 TS	All-336 TS	Gain	Aligned tokens
EN→DE	90.40	68.49	+21.91	11,209
EN→ZH	93.48	65.79	+27.70	7,582
EN→IT	91.90	67.42	+24.49	12,056

Table 3: **MT head-set filtering ablation on held-out word-aligned dev examples.** To make the all-head baseline maximally charitable, we average the candidate head set first and then restrict the argmax to the source-prompt token zone before scoring against gold aligned source tokens. Scores are reported in points ($100 \times \text{TS}$). Even under this source-only argmax, the per-direction retained top-8 heads still preserve a markedly stronger alignment signal than uniform all-head averaging.

sliding-window and suffix-side causal/window masks, we compute

$$\mathbf{P}^{(\ell,h)} = \gamma^{(\ell)} \mathbf{B}_{\text{dQ}}^{(\ell,h)} (\mathbf{B}_{\text{pK}}^{(\ell,h)})^\top + \mathbf{M}_{\text{prompt}}^{(\ell,h)}, \quad (9)$$

$$\mathbf{S}^{(\ell,h)} = \gamma^{(\ell)} \mathbf{B}_{\text{dQ}}^{(\ell,h)} (\mathbf{B}_{\text{dK}}^{(\ell,h)})^\top + \mathbf{M}_{\text{suffix}}^{(\ell,h)}, \quad (10)$$

$$\tilde{\mathbf{A}}^{(\ell,h)} = \text{softmax}[\mathbf{P}^{(\ell,h)}, \mathbf{S}^{(\ell,h)}]. \quad (11)$$

Restricting $\tilde{\mathbf{A}}^{(\ell,h)}$ to the source columns $\phi^{(k)}(s)$ recovers the policy-visible block of Eq. (2). The replay cost therefore scales with the selected head set and the short draft length, not with the full LHn^2 attention tensor.

Qualitative reconstruction example. Figure 9 shows the full reconstruction stack on an EN→DE text-only MT probe: the exact prompt partition seen by the decoder, word-level aggregation of the replayed rows for readability, and the four-way provenance accounting for every drafted word.

The green band in Figure 9 is previously committed target text reused as causal context; it is not

itself the object of the gate. The orange words are the current draft scanned left-to-right by the policy. We allow $\hat{s}_t$ to move backward within a live draft when the decoder revisits earlier source material under reordering, which is why the residual prompt and suffix masses remain explicit in the decoder-only formulation.

Numerical parity. Let $\mathbf{A}^{(\text{TF})}$ denote the reference attention tensor produced by an uncompiled Transformers forward on the identical prompt. On a curated parity set, we observe

$$\begin{aligned} \|\Delta \mathbf{A}\|_\infty &\leq 1.2 \times 10^{-2}, \\ \|\Delta \mathbf{A}\|_1 / (nL_k) &\leq 4 \times 10^{-4}, \end{aligned} \quad (12)$$

with bit-identical acceptance decisions between the eager and compiled paths. The observer is therefore not just fast enough to keep, but faithful enough to support controlled MT ablations.

Word-level selective reconstruction on a live MT draft

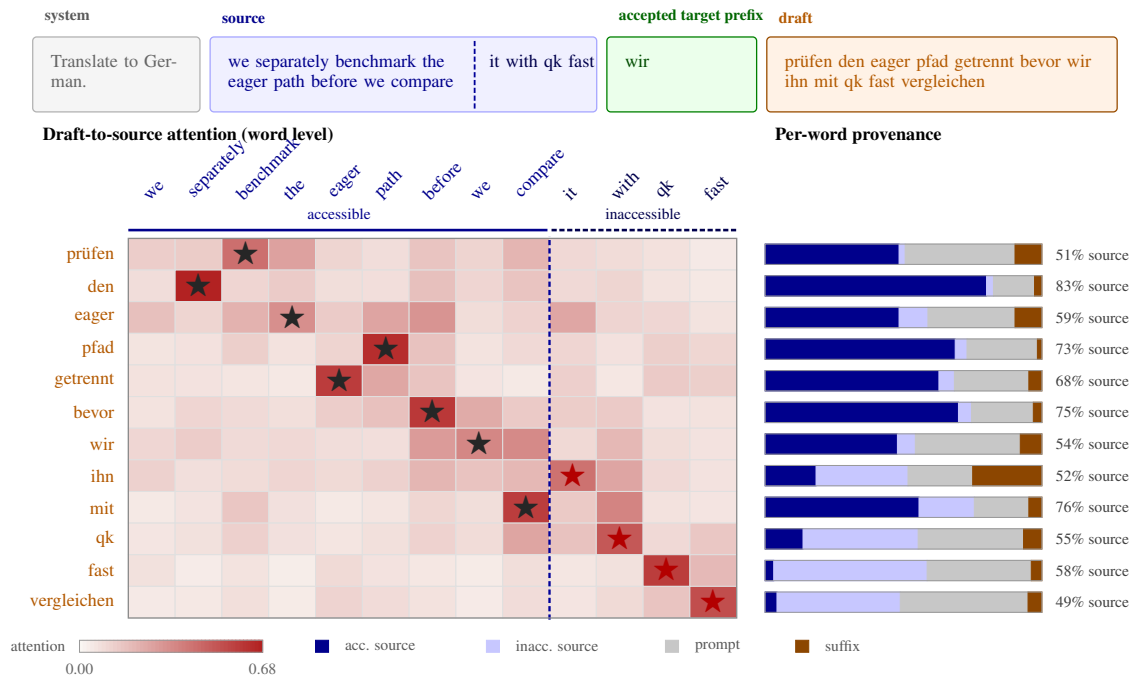


Figure 9: **Word-level selective reconstruction on a live MT draft.** *Top ribbon:* prompt partition into system instruction, live source, accepted target prefix, and current draft. *Left panel:* reconstructed draft-to-source attention from the selected AlignAtt heads, aggregated to words and split by the dashed accessibility frontier; black stars stay on the accessible side, while the first red star marks the SOURCE-FRONTIER failure. *Right panel:* each drafted word is decomposed into accessible-source, inaccessible-source, prompt, and suffix mass, with the numeric label giving the total source share.